

SIMATS ENGINEERING

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

ASSESSMENT TOOL 1

:

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I J.Vishnu Vardhan(192324297) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Application-Based Questions

Q3. Keyword-Based Encryption (Vigenère Cipher)

Given: Plaintext = HELLO, Keyword = KEY

a. Step-by-step encryption

Step 1: Convert letters to numbers (A=0 ... Z=25)

H=7, E=4, L=11, L=11, O=14

Step 2: Repeat the keyword to match the message length

	H	E	L	L	O
Key (repeated)	K	E	Y	K	E
Key value	10	4	24	10	4

Step 3: Apply $C = (P + K) \bmod 26$

Letter	P + K	mod 26	Cipher
H	$7 + 10 = 17$	17	R
E	$4 + 4 = 8$	8	I
L	$11 + 24 = 35$	$35 - 26 = 9$	J
L	$11 + 10 = 21$	21	V
O	$14 + 4 = 18$	18	S

Ciphertext = RIJVS

Decryption check: $P = (C - K) \bmod 26$. For example, $R(17) - K(10) = 7 \rightarrow H$ ✓.

b. Influence of the repeating keyword

- Each plaintext letter is shifted by a different amount, determined by the key letter aligned with it (K = 10, E = 4, Y = 24).
- The shifts cycle $K \rightarrow E \rightarrow Y \rightarrow K \rightarrow E$, so the key repeats every 3 letters.
- **The same plaintext letter encrypts differently.** The two L's become J and V because they align with different key letters (Y and K). This hides letter frequency, which a simple Caesar cipher does not.
- Because the key repeats, patterns recur every 3 positions. This is the weakness exploited by the Kasiski examination and the Index of Coincidence.

Q5. Legacy Transposition Cipher (Rail Fence, 3 Rails)

Ciphertext: WECRLTEERDSOEEFEAOCAIVDEN (25 letters)

a. Reversing the encryption

Step 1: Work out the rail lengths. In a 3-rail zigzag of 25 characters, the pattern repeats every 4 positions (0-1-2-1):

- Rail 1 (positions 0, 4, 8, ...) has **7** letters
- Rail 2 (positions 1, 3, 5, ...) has **12** letters
- Rail 3 (positions 2, 6, 10, ...) has **6** letters

Step 2: Split the ciphertext by rail

- Rail 1: **WECRLTE**
- Rail 2: **ERDSOEEFEAOC**
- Rail 3: **AIVDEN**

Step 3: Place the letters back in the zigzag

Rail 1: W . . . E . . . C . . . R . . . L . . . T . . . E
Rail 2: . E . R . D . S . O . E . E . F . E . A . O . C .
Rail 3: . . A . . I . . V . . D . . E . . N . .

Step 4: Read column by column (zigzag order)

WEAREDISCOVEREDFLEEATONCE

Original message: "WE ARE DISCOVERED. FLEE AT ONCE."

b. Why this method is vulnerable today

- **Tiny key space.** The only key is the number of rails (2 to about $n-1$), so brute force tries every option in milliseconds.
- **Letter frequencies are unchanged.** Transposition only rearranges letters, so the statistics of the plaintext survive. Frequency analysis shows it is not a substitution cipher.
- **Weak to anagramming and known-plaintext attacks.** Common words or phrases ("THE", "FLEE", "ATTACK") are easy to spot and test.
- **No diffusion or confusion.** A change of one character affects only one position in the ciphertext, unlike modern ciphers such as AES.
- **No integrity or authentication.** Messages can be altered or replayed without detection.
- **Computers solve it automatically.** Pattern-based software can crack it almost instantly.

c. Practical enhancement

Use a product cipher: substitution followed by transposition. First encrypt with a keyed substitution (for example Vigenère with a long random key, as in Q3), then apply the rail-fence transposition. This gives confusion (substitution) and diffusion (transposition), so frequency analysis no longer works directly.

For real confidentiality, the best option is to replace the legacy method with **AES-256 (GCM mode)**, which provides strong encryption and integrity checking.